

MOMBASA INTL, Kenya

WMO: 638200

Lat: 4.035S

Lon: 39.594E

Elev: 61

StdP: 100.59

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	20.9	17.7	12.7	27.0	18.3	13.3	26.7	10.4	27.2	9.6	26.8	1.9	210	0.367

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	7.4	33.2	25.3	32.8	25.2	32.2	25.1	26.6	30.3	26.2	29.8	26.1	29.7	5.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	21.1	28.2	25.2	20.5	27.8	25.1	20.4	27.7	83.4	30.6	82.1	30.3	81.0	30.2	33.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.2	8.3	7.4	DB	17.7	35.5	3.5	1.3	15.2	36.4	13.1	37.1	11.2	37.9	8.7	38.8	(4)
(5)			WB	16.7	28.4	3.1	1.6	14.4	29.5	12.6	30.5	10.8	31.3	8.5	32.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.8	28.1	28.4	28.8	28.1	26.5	25.5	24.7	24.6	25.3	26.3	27.2	28.0	(6)
(7)		DBStd	1.70	0.75	0.80	0.80	1.15	1.11	0.81	0.75	1.00	0.75	0.84	1.00	0.79	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6123	562	516	584	542	510	464	455	453	459	506	515	558	(10)
(11)		CDD18.3	3082	303	282	326	292	252	214	197	194	209	248	265	299	(11)
(12)		CDH23.3	27952	3065	3004	3804	3161	2049	1442	1116	1128	1429	2071	2588	3094	(12)
(13)	CDH26.7	8754	1143	1166	1544	1125	518	283	164	162	275	501	764	1110	(13)	
(14)	Wind	WSAvg	3.4	3.7	3.4	3.0	3.5	3.9	3.8	3.7	3.5	3.5	3.2	2.5	2.8	(14)
(15)	Precipitation	PrecAvg	1085	29	15	49	153	241	109	70	66	56	116	108	63	(15)
(16)		PrecMax	2201	224	83	146	321	507	217	153	181	161	769	290	261	(16)
(17)		PrecMin	676	0	0	0	30	21	16	14	27	2	0	0	0	(17)
(18)		PrecStd	364	47	22	39	90	114	54	34	32	44	153	71	58	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	33.9	34.1	33.2	32.0	30.2	29.7	29.2	30.1	31.1	32.1	33.2	(19)
(20)			MCWB	24.8	25.0	25.5	25.6	24.2	23.4	22.9	22.5	23.1	24.3	25.5	25.5	(20)
(21)		2%	DB	32.8	33.0	33.1	32.8	31.0	29.5	28.9	28.9	29.2	30.2	31.2	32.2	(21)
(22)			MCWB	24.8	25.0	25.5	25.5	24.5	23.2	22.4	22.4	22.9	24.1	25.3	25.4	(22)
(23)		5%	DB	32.1	32.2	32.8	32.0	30.1	29.0	28.1	28.1	28.9	29.9	30.9	31.9	(23)
(24)			MCWB	24.6	24.8	25.4	25.3	24.4	23.1	22.3	22.3	22.8	24.1	25.3	25.3	(24)
(25)		10%	DB	31.3	31.9	32.1	31.1	29.3	28.2	27.8	27.8	28.2	29.1	30.1	31.1	(25)
(26)			MCWB	24.6	24.7	25.3	25.3	24.3	23.1	22.3	22.3	22.7	23.8	25.1	25.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.2	26.5	27.1	26.7	26.1	25.1	24.1	24.1	24.7	25.8	26.6	27.0	(27)
(28)			MCD B	30.5	31.3	30.9	30.2	29.2	27.9	27.0	26.7	27.6	28.7	30.1	30.9	(28)
(29)		2%	WB	25.7	26.0	26.6	26.4	25.6	24.6	23.6	23.5	24.1	25.3	26.1	26.4	(29)
(30)			MCD B	30.0	30.4	30.3	29.8	28.6	27.5	26.6	26.3	27.0	28.1	29.4	30.1	(30)
(31)		5%	WB	25.4	25.6	26.2	26.1	25.2	24.2	23.2	23.2	23.7	24.9	25.7	26.1	(31)
(32)			MCD B	29.6	29.9	29.9	29.5	28.2	27.1	26.2	26.1	26.6	27.6	28.9	29.7	(32)
(33)		10%	WB	25.1	25.2	26.0	25.9	24.9	23.8	22.9	22.8	23.4	24.6	25.5	25.8	(33)
(34)			MCD B	29.3	29.4	29.6	29.2	27.8	26.7	25.8	25.7	26.4	27.3	28.6	29.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.0	7.9	7.4	6.6	6.0	6.3	6.6	6.7	6.6	6.3	6.5	7.3	(35)
(36)			MCD BR	8.3	8.2	7.9	7.3	7.0	6.9	7.1	7.2	7.2	6.9	7.0	7.8	(36)
(37)			MCWBR	2.3	2.2	2.3	2.2	2.5	2.6	2.6	2.5	2.4	2.3	2.4	2.4	(37)
(38)		5% WB	MCD BR	7.9	7.9	7.3	6.8	6.1	6.2	6.5	6.5	6.4	6.0	6.5	7.2	(38)
(39)			MCWBR	2.4	2.4	2.4	2.3	2.6	2.7	2.6	2.6	2.5	2.4	2.5	2.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.482	0.490	0.450	0.411	0.422	0.442	0.441	0.425	0.414	0.415	0.416	0.460	(40)	
(41)		taud	2.222	2.193	2.355	2.497	2.445	2.343	2.325	2.374	2.400	2.419	2.440	2.317	(41)	
(42)		Ebn at Noon	859	853	878	889	852	820	828	864	897	909	913	875	(42)	
(43)		Edh at Noon	150	155	131	109	111	120	123	122	123	122	120	135	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.98	6.24	6.09	5.39	4.71	4.63	4.77	5.14	5.76	6.00	6.02	5.98	(44)	
(45)		RadStd	0.29	0.22	0.33	0.32	0.40	0.22	0.22	0.22	0.26	0.28	0.31	0.25	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.36	+0.47	+0.46	+0.31	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.36	+0.47	+0.46	+0.31	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page